DSA Daily Quiz 5

Total points 14/15

The respondent's email (**omjbhaleraowork@gmail.com**) was recorded on submission of this form.

Full Name *

Om Jairam Bhalerao

Binary search algorithm cannot be applied to ... * 1/1

- (•) sorted linked list ✓
- ( ) sorted binary trees
- ( ) sorted linear array
- ( ) pointer array

For a sorted list the best sorting algorithm is * 1/1

- ( ) Quick sort
- ( ) Insertion sort
- ( ) Selection sort
- (•) Bubble sort ✓

✓ Which of the following is not a stable sorting algorithm? * 1/1

- ○ Insertion sort
- ◉ Selection sort ✓
- ○ Bubble sort
- ○ Merge sort

✓ What is an external sorting algorithm? * 1/1

- ◉ Algorithm that uses tape or disk during the sort ✓
- ○ Algorithm that uses main memory during the sort
- ○ Algorithm that involves swapping
- ○ Algorithm that are considered 'in place'

✓ Given an unsorted array. The array has this property that every element in the array is at most k distance from its position in a sorted array where k is a positive integer smaller than the size of an array. Which sorting algorithm can be easily modified for sorting this array and what is the obtainable time complexity? * 1/1

- ○ Insertion Sort with time complexity O(kn)
- ◉ Heap Sort with time complexity O(nLogk) ✓
- ○ Quick Sort with time complexity O(kLogk)
- ○ Merge Sort with time complexity O(kLogk)

Which of the following algorithm pays the least attention to the ordering of the elements in the input list? * 1/1

- Insertion sort
- Selection sort
- Quick sort
- None

Which of the following is not an in-place sorting algorithm? * 1/1

- Selection sort
- Heap sort
- Quick sort
- Merge sort

Which of the following sorting algorithms has the minimum worst-case time complexity? * 1/1

- Quick sort
- Selection sort
- Bubble sort
- Merge sort

If the number of records to be sorted is small, then ...... sorting can be efficient. *0/1

- ( ) Merge
- ( ) Heap
- ( ) Selection
- (•) Bubble ✗

Correct answer

- (•) Selection

Sort the following list using the Selection Sort algorithm. * 1/1
H, V, A, T, L, N, K
What is the output of the algorithm after the third pass?

- ( ) A, V, T, H, L, N, K
- ( ) A, T, H, K, L, N, V
- ( ) A, V, H, T, L, N, K
- (•) A, H, K, T, L, N, V ✓

Which of the following sorting procedures is the slowest? * 1/1

- ( ) Quick sort
- ( ) Heap sort
- ( ) Shell sort
- (•) Bubble sort ✓

To merge 4 sorted files having 10,10,10 and 10 records. How much time it will take? *1/1

- O(100)
- O(64)
- O(40)
- O(10000)

Which of the following sorting techniques follows the divide-and-conquer approach? *1/1

- Bubble sort
- Selection sort
- Insertion sort
- Merge sort

Last TWO Digits of the PRN *

48

What is the worst case complexity of bubble sort? * 1/1

- O(nlogn)
- O(logn)
- O(n)
- O(n2)

What is the advantage of bubble sort over other sorting techniques? * 1/1

- It is faster
- Consumes less memory
- Detects whether the input is already sorted
- All of the mentioned